

# Errors in Variables

Every earlier chapter takes the regressors as recorded without error. Often they are not: one clinic reading stands in for long-run blood pressure, a questionnaire for usual diet. The scientific model concerns the true value  $X$ , but the data contain a noisy  $W$ , and least squares on  $W$  estimates the best linear predictor from what was recorded. This chapter measures the gap and closes it. Classical error flattens a slope by a factor no sample size removes, makes correctly measured covariates look important, and in the normal model leaves the slope unidentified. Information from outside the regression rescues the analysis: a known error variance, repeated measurements, a known variance ratio, an instrument, or validation data. The moment correction, instrumental variables, regression calibration and SIMEX turn that information into estimates, each with its own conditions and costs.

**What you need.** [Chapter 6](#) (the population projection, Theorem 6.11.1 and Proposition 6.11.3, and Theorem 6.3.6). [Chapter 5](#) (moments of least squares under departures, Proposition 5.5.7). [Chapter 14](#) (random regressors, Theorem 14.1.2, and the limit theorems of [Section 14.4](#)). From this part, [Chapter 19](#) (omitted-variable bias, Theorem 19.1.2), [Chapter 21](#) (sandwich covariance estimators, Theorem 21.4.2) and [Chapter 23](#) (the pairs bootstrap, Definition 23.1.1). The measurement-error model itself is a statement about moments, so [Chapter 2](#) (Theorem 2.2.1, Theorem 2.5.2) and [Chapter 3](#) (Theorem 3.3.1, Theorem 3.5.2) are used throughout [Sections 24.2 to 24.5](#), with the partitioned inverse and the extremal Rayleigh quotient of [Chapter 1](#) (Theorem 1.6.4, Theorem 1.7.7) and, in [Section 24.4](#), the warning about a ratio with no mean from [Chapter 12](#) (Proposition 12.5.2).

## 24.-1.1 Roadmap

- [Attenuation bias](#)
- [The classical measurement error model](#)
- [Reliability and moment correction](#)
- [Instrumental variables](#)
- [Regression calibration and SIMEX](#)
- [Summary and notes](#)